

CII-SR Kaizen Circle Competition 2018

A Kaizen Presentation By

(EVEREST INDUSTRIES LIMITED)

On: 26th July-18

Venue : Hotel Westin, Chennai

Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing, ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of designs, drawings and implementation for every project.

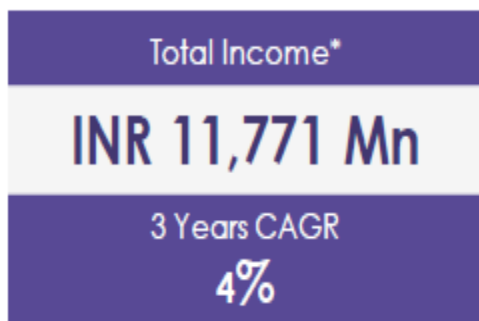
FY17 Business Mix (Revenue Share %):

- **Building products (63%)** – includes fibre cement roofing sheets, fibre cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customised building solutions like Pre-Engineered Steel Buildings, Smart Steel Buildings, metal roofing sheets and cladding.

Manufacturing and Distribution Networks:

- 6 Building Products plants and 3 Steel Building Plants.
- 40 Sales Depot, over 6,000 Dealer Outlets, serving over 600 cities & 100,000 villages.
- Export to over 35 countries (Green solutions – Fibre cement boards).

FY17 Financial Snapshot (Consolidated):



*Total income includes other income

Everest range of products includes



**Fibre Cement Roofing
Sheets & Accessories**



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings





Kaizen Theme: To Reduce power consumption by eliminating vacuum pump & 60hp motor.

Implemented Area: Sheeting Machine -1 Shop floor, Podanur Works, Coimbatore

Cross Functional Team:



Mr. M. Palavesa Moorthy
Shift Incharge



Mr. M. Madan Raj
Shift Incharge



Team Leader
Mr. Y. Kesavaraj
Line Incharge



Mr. K. Senthil Kumar
Electrical Supervisor



Mr. Sivadasan
Machine Driver

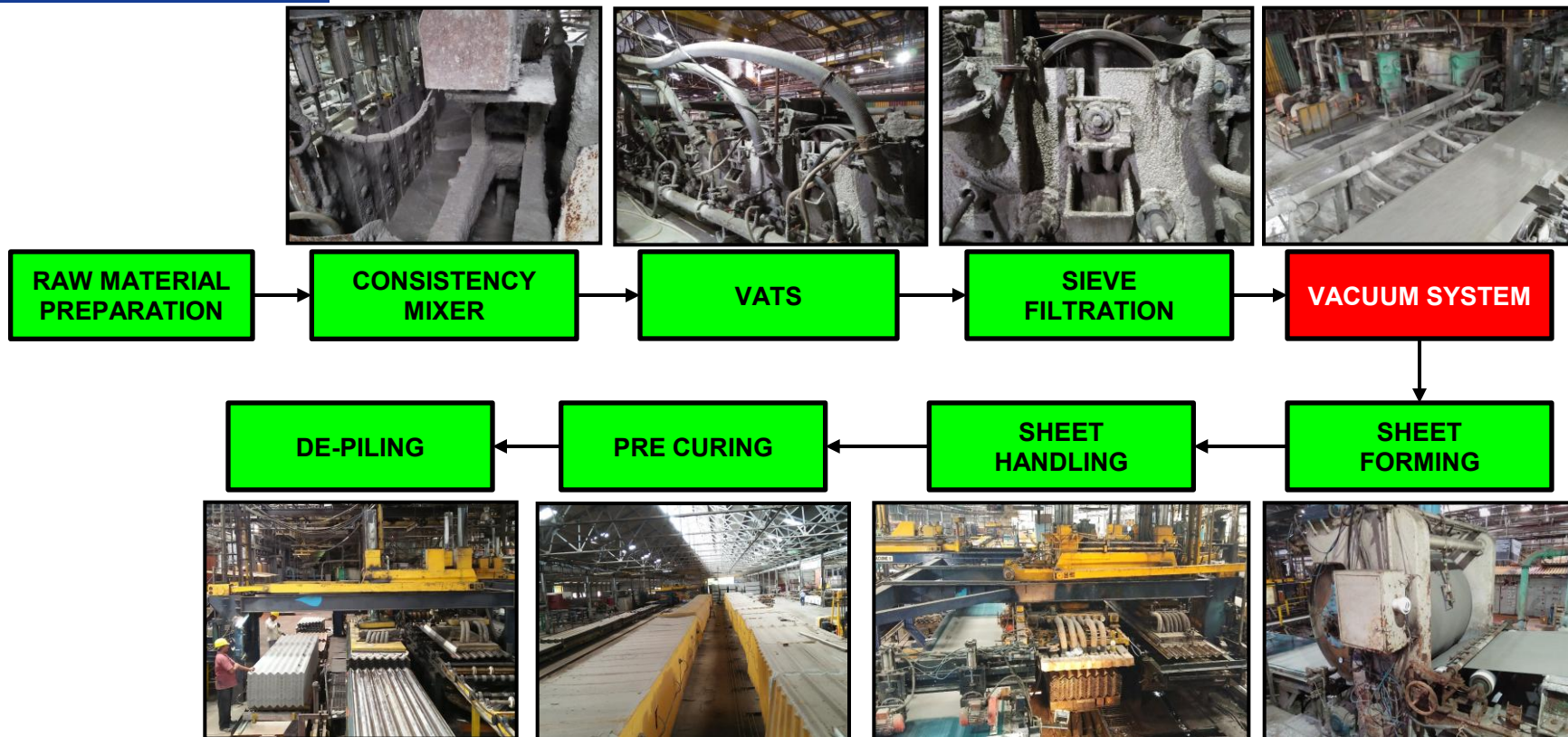


Mr. Madhialagan
Machine Driver



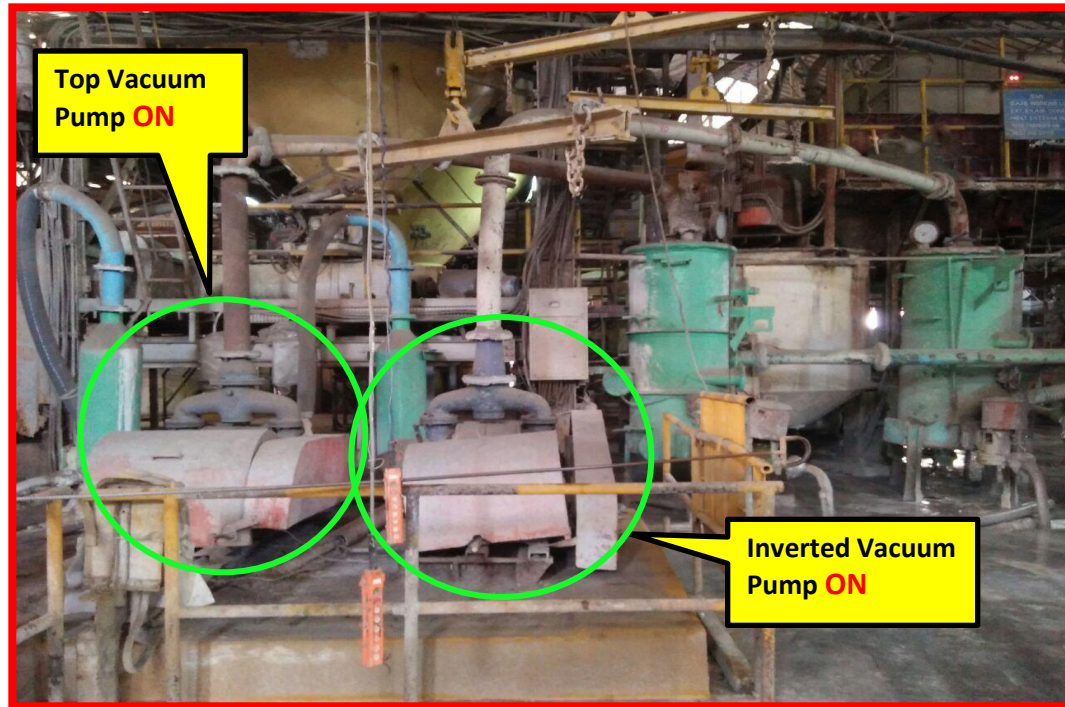
Mr. Peter
Machine Driver

Process Flow:



Problem:

More power consumption due to higher vacuum. If we reduce Vacuum below the optimum requirement we are facing problems like Layer falling from felt, Layer transferring problem to felt & Vat over concentration.



BOTTOM VACUUM TRAY



TOP VACUUM TRAY

- Bottom suction boxes openings are high and using more no. of Top suction boxes with rollers arrangement for decreasing the load on the Main drive due to high vacuum.



- In this, it shows flow of Vacuum through the pipe lines & vacuum chambers towards Top & Inverted vacuum pumps when both the vacuum pumps are in **ON** Condition.

EVEREST Root Cause Analysis & Idea Generation

Problem: More power consumption due to higher vacuum usage.

Why 1 : More Vacuum is used due to high volume.

Why 2 : Suction boxes openings are high & intensity of vacuum is less, for that we were running 2 vacuum pumps to hold the film.

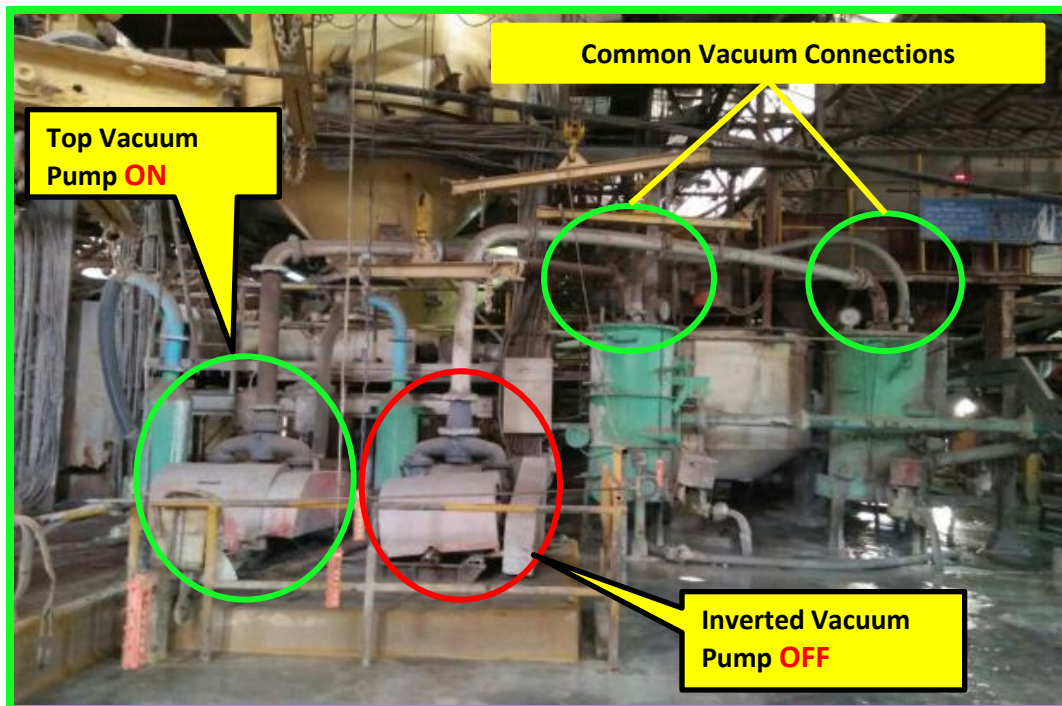
Why 3 : Design Constraint.

Root Cause: Design constraint.

Idea Generation: Modifying the design of the vacuum suction arrangement.

Action: Modifying the design of the suction boxes opening & removed additional suction boxes from the circuit. And providing common vacuum line b/w vacuum chambers.

SN	Counter Measure Activity
01	Modified the opening area of Bottom vacuum trays.
02	Removed rollers from Top vacuum trays and installed Polypicks to provide more intensity of vacuum for film drying faster & removed extra Top vacuum suction trays by reducing the volume of vacuum.
03	Provided common vacuum line b/w top trays vacuum chamber & Bottom vacuum chamber to utilize one vacuum pump instead of two vacuum pumps.



BOTTOM VACUUM TRAY



TOP VACUUM TRAY

- By modifying the vacuum circuit, now only 1 vacuum pump using for Top and Bottom trays instead of using Two vacuum pumps.
- By this we have stopped using 1 Vacuum pump, 60 Hp motor, Rollers & Bearings in vacuum trays & other consumables for maintenance.



- In this, you can see Top vacuum pump is **ON** bottom vacuum pump is **OFF**.
- Vacuum flows towards Top Vacuum Pump through Top & Inverted vacuum trays pipe lines & vacuum chambers. But vacuum does not flows towards Inverted vacuum pump.
- Vacuum from inverted vacuum trays flows towards Top vacuum pump through pipe lines, then to Inverted vacuum chamber and then to Top vacuum chamber.

Before Situation:

Vacuum Trays opening Area

Area = 2044 CM²

Bottom Vacuum Trays

Area = 1898 CM²

Top Vacuum Trays

After Situation:

Area = 1168 CM²

Bottom Vacuum Trays

Area = 2920 CM²

Top Vacuum Trays

Calculation of Power consumption & Cost Savings of Vacuum Pumps:

- Previously we are using 2 Vacuum Pumps of 60Hp & 1 Vacuum Pump of 40Hp.
- Now we are using 1 Vacuum Pump of 60Hp & 1 Vacuum Pump of 40Hp.

$$\begin{aligned}\text{Power Saving} &= \sqrt{3} \times V \times I \times \cos\phi \\ &= (\sqrt{3} \times 415 \times 70 \times 0.8) / 1000 \\ &= 34.502 \times 24 \text{ Hrs} \times 28 \text{ Production days} \\ &= \mathbf{27,049 \text{ Units/Month By eliminating 60Hp Vacuum Pump motor}}\end{aligned}$$

$$\begin{aligned}\text{Cost Saving per Month} &= \mathbf{27,049 \times 8 \text{ Rs. Per Unit}} \\ &= \mathbf{\text{Rs. 2,16,392/-}}\end{aligned}$$

$$\text{Cost Saving per Year} = \mathbf{25.96 \text{ Lakh/Annum}}$$

- Cost Saving by eliminating Roller suction trays & Bearings = **1.72 Lakh/Annum**
- Cost Saving by eliminating Maintenance of 1 Vacuum Pump = **Rs. 17,000/Annum**
- **Total Cost Saving is 27.95 Lakh/Annum**

Quantitative : Potential Cost Saving of 27.95 Lakh/Annum.

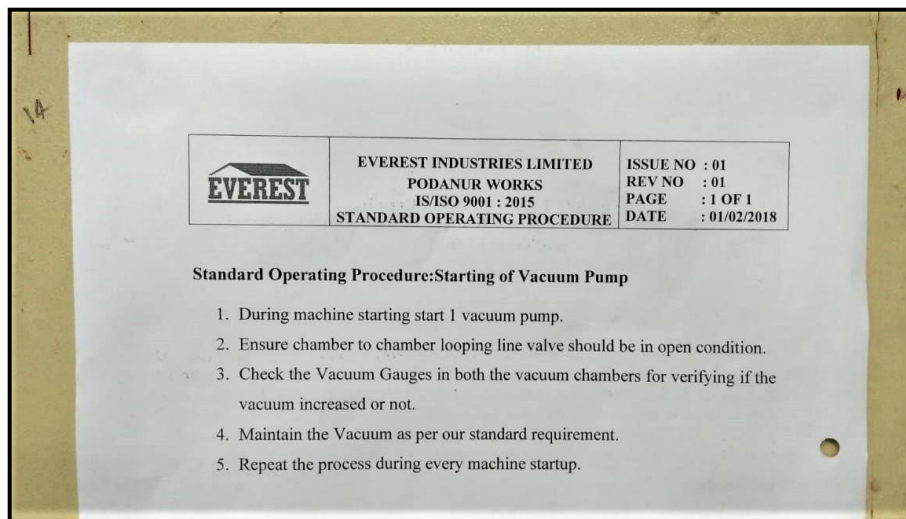
Energy : Average monthly Power consumption of vacuum pump motors is reduced from 71486 Units to 44437 Units, which is 37.83% reduction in Vacuum pumps power consumption.

Cost : Potential Cost Savings = 27.95 Lakh/Annum

Safety : This system does not require any intervention by the people.
Hence the system is safe.

Morale : It takes less effort for maintenance of vacuum pumps & power consumption is reduced. By this morale of employees increased.

- Training on Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.



Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Scope of Horizontal Deployment in across Everest :-

Factory	Target Date	Status
1.Bhagwanpur Factory, Roorkee	31.07.2018	Under Implementation
2.Kolkata Factory, Kolkata	30.07.2018	Under Implementation
3.Kymore Factory, Madhya Pradesh	15.08.2018	Under Implementation
4.Podanur Factory, Coimbatore	31.03.2018	Implemented
5.Somnathpur Factory, Odisha	30.07.2018	Under Implementation

Kaizen Sheet



Company

MM/YY

S.No

Productivity

Quality

Delivery

Safety

Material

Energy

Maintenance

Everest Industries Ltd

03/18

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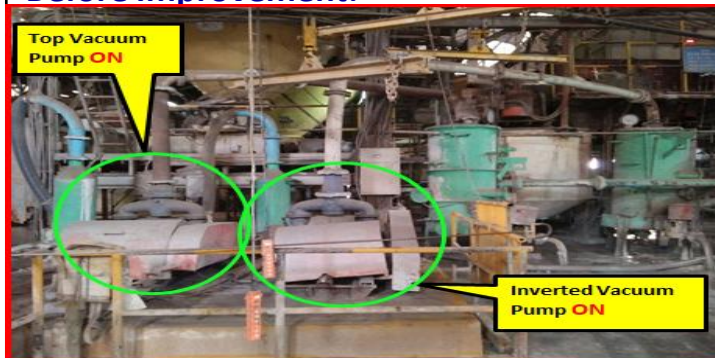
Kaizen Title: Reducing power consumption by eliminating vacuum pump & 60hp motor.

Implemented Area: Shop floor

Problem/Present Status:

More power consumption and due to higher vacuum. If we reduce Vacuum below the optimum requirement we are facing problems like Layer falling from felt, layer transferring problem to felt & vat over concentration problem.

Before Improvement:



Implemented by:

Y. Kesavaraj (TL), M. Palavesa Moorthy, M. Madan Raj, K. Senthil Kumar, N. Anil Sivadasan, Peter, Madhialagan

Result/Benefit:

(a) Qualitative

We achieved our Quality bench mark.

Root Cause Identification:



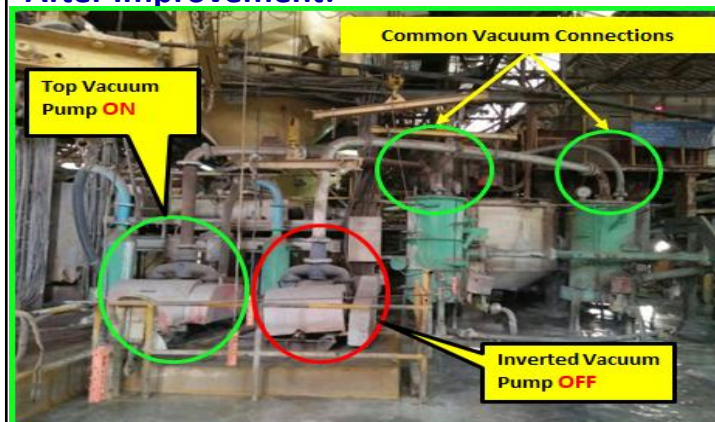
Problem
•More power consumption due to higher vacuum usage.

Why 1
•More Vacuum.

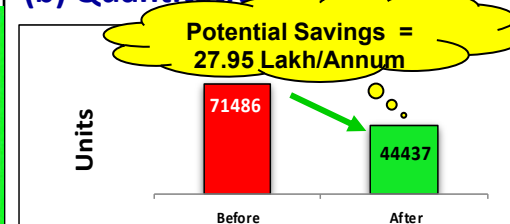
Why 2
•Suction box openings are high & intensity of vacuum is less, for that we were running 2 Vacuum pumps to hold the film

Why 3
•Design Constraint

After Improvement:



(b) Quantitative



Monthly Avg. Power Consumption of Vacuum Pumps

Standardization:

Preventive maintenance as per schedule.

Root cause: Design constraint.

Idea to eliminate root cause: Modifying the suction box opening & removed additional suction boxes from the circuit. And providing common vacuum line b/w vacuum chambers.

Counter-measure: We modified the suction boxes opening as per requirement of the film thickness, removed one additional Top suction box and provided one guide roller in that place. And by connecting common vacuum line from vacuum chamber to chamber we switched OFF 1 Vacuum pump with motor of 60 hp.

Horizontal Deployment:

- 1) Bhagwanpur Unit - Under Implementation
- 2) Kolkata Unit – Under Implementation
- 3) Kymore Unit – Under Implementation
- 4) Podanur Unit – Implemented
- 5) Somnathpur Unit – Under Implementation